

K Bhaskar Chari

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PROFESSIONAL SUMMARY

AI/ML Engineer focused on bridging the gap between research and production. Expertise in architecting grounded RAG systems and scalable inference pipelines, translating complex model capabilities into reliable, low-latency software solutions.

MACHINE LEARNING PROJECTS

DocuMind AI

- **RAG Architecture:** Engineered a FastAPI + Qdrant pipeline with Semantic Chunking, reducing inference latency by 20% using SentenceTransformers (all-mpnet-base-v2).
- **Safety & Validation:** Built a custom Validation Layer and Confidence Scoring Engine for Gemini, cutting hallucinations by 35% and enforcing strict factual adherence.
- **Reliability:** Deployed Fault-Tolerant async endpoints with 100% Traceability, ensuring every generated answer links directly back to the specific source page.

TarangAI

- **Computer Vision:** Developed a multi-output Neural Network using MediaPipe for real-time pose extraction, achieving 89% classification accuracy across complex Indian classical dance styles.
- **GenAI Integration:** Implemented a Pose Refinement GAN for real-time pose correction, improving pose recognition quality by 25% and enhancing feedback accuracy on posture, mudras, and facial expressions.

Crop Yield Prediction Using ANN

- **Predictive Modeling:** Tailored an ANN-based forecasting model integrating multi-factor environmental data, outperforming traditional ML baselines with a 15% accuracy improvement.
- **Pipeline Optimization:** Designed a complete ML training pipeline with data preprocessing, model training, and hyperparameter tuning, reducing training time by 20% and overall compute cost by 10%.

ENGINEERING EXPERIENCE

India Space Lab (ISL)

India

Winter Technical Intern – Space Science & Technology

Dec 2024 – Jan 2025

- Completed hands-on technical training in space science and aerospace systems, gaining practical exposure to mission planning, payload design, and space-grade constraints.
- Analyzed small satellite and UAV system architectures through applied CanSat, CubeSat, and advanced drone modules, strengthening understanding of aerospace workflows and mission-level design trade-offs.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning, NLP, Computer Vision, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering

ML Systems & MLOps: End-to-end ML pipelines, experiment tracking, model versioning, inference optimization, GitHub Actions

Frameworks & Libraries: PyTorch, TensorFlow, LangChain, Scikit-Learn, FAISS, MediaPipe, CatBoost, NLTK, Pandas, NumPy, PyTest, Matplotlib

Deployment & Tooling: FastAPI, REST APIs, Docker/Containerization, Git, GitHub, Jupyter Notebook

Databases & Data: MySQL, Qdrant (vector database)

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science Engineering (AIML) — CGPA: 8.1

Bhopal, Madhya Pradesh

Oct 2022 – May 2026 (Expected)